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Module 1: Introduction to Machine Learning

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Introductions

- Tell us about you and:
  - Your research/professional interests
  - How do you anticipate Machine Learning can assist you

Learning Objectives

- Understand the concept of supervised and unsupervised Machine Learning (ML).
- Understand what ML can and cannot do.
- Understand how to measure performance of ML models.
- Understand the concepts of overfitting and underfitting.

Supervised and Unsupervised Learning

- Essentially, every Machine Learning (ML) application can be tagged either as **Supervised Learning** or **Unsupervised Learning**.
- These are two fundamental concepts that are defined by how ML uses its data.

Supervised Learning

- Supervising learning is defined by the use of labelled data.
- You know beforehand which are the features **X** that describe a target $\mathbf{y}$.
- Supervision comes from observing $\mathbf{y}$ during training.
- The algorithm learns the relationship between **X** and $\mathbf{y}$.
- Then, it attempts to generalize the learned pattern to unseen **X.**

Supervised Learning

- Two main families of supervised learning.
- **Classification:** prediction of discrete values (classes)
- **Regression:** prediction of continuous values (numbers)

Supervised Learning

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<u>Problem</u>: Predict whether a patient will survive 30 days

Supervised Learning

- Two main families of supervised learning.
- **Classification:** prediction of discrete values (classes)
- **Regression:** prediction of continuous values (numbers)

<u>Problem</u>: Predict how many days a patient will survive

Supervised Learning

- Two main families of supervised learning.
- **Classification:** prediction of discrete values (classes)
- **Regression:** prediction of continuous values (numbers)

<u>Problem</u>: Predict a patient's systolic blood pressure

Supervised Learning

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<u>Problem</u>: Predict cancer stage I, II, III, or IV.

Supervised Learning

- Two main families of supervised learning.
- **Classification:** prediction of discrete values (classes)
- **Regression:** prediction of continuous values (numbers)

Problem: Predict pain-score (0-10)

Example: Supervised Learning, Classification

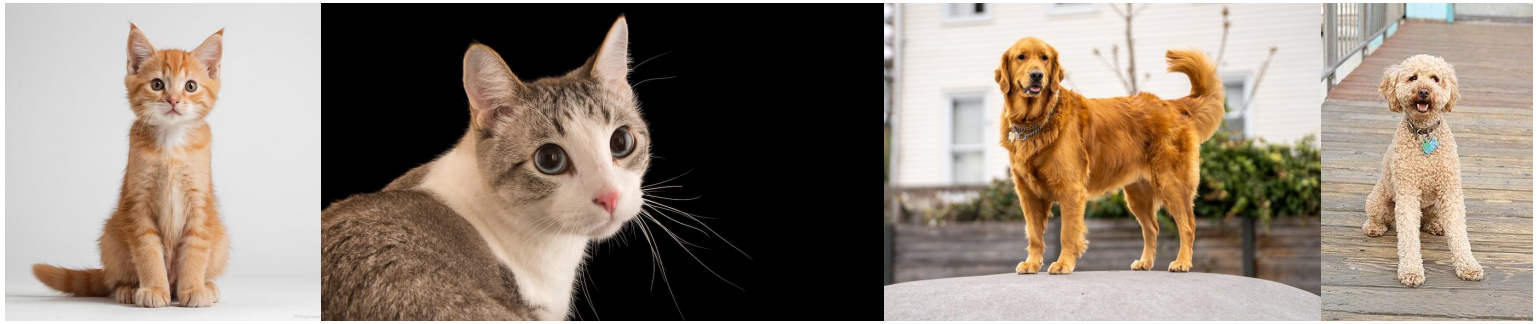
- **y** (binary; cat or dog; 1/0)

Cat | Dog

- **X** (features that describe y)

Ear shape

Training: We analyze n pictures of cats and dogs and try to find a ear shape that describes cats and dogs.

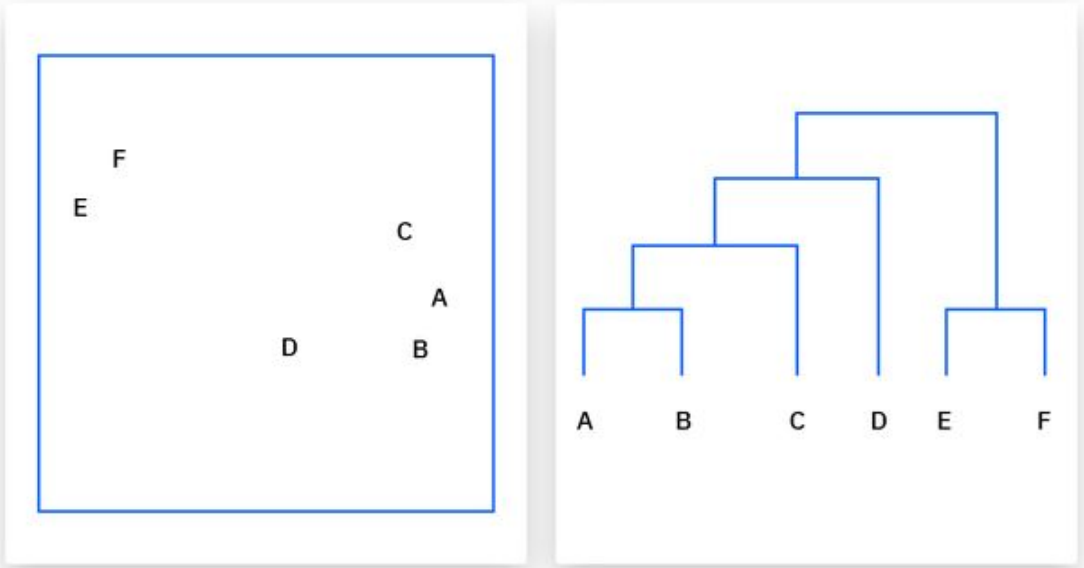


Unsupervised Learning

- Analyze unlabeled data sets to discover hidden patterns.
- There is no target label (**y**).
- Generally requires the involvement of someone familiar with the data.

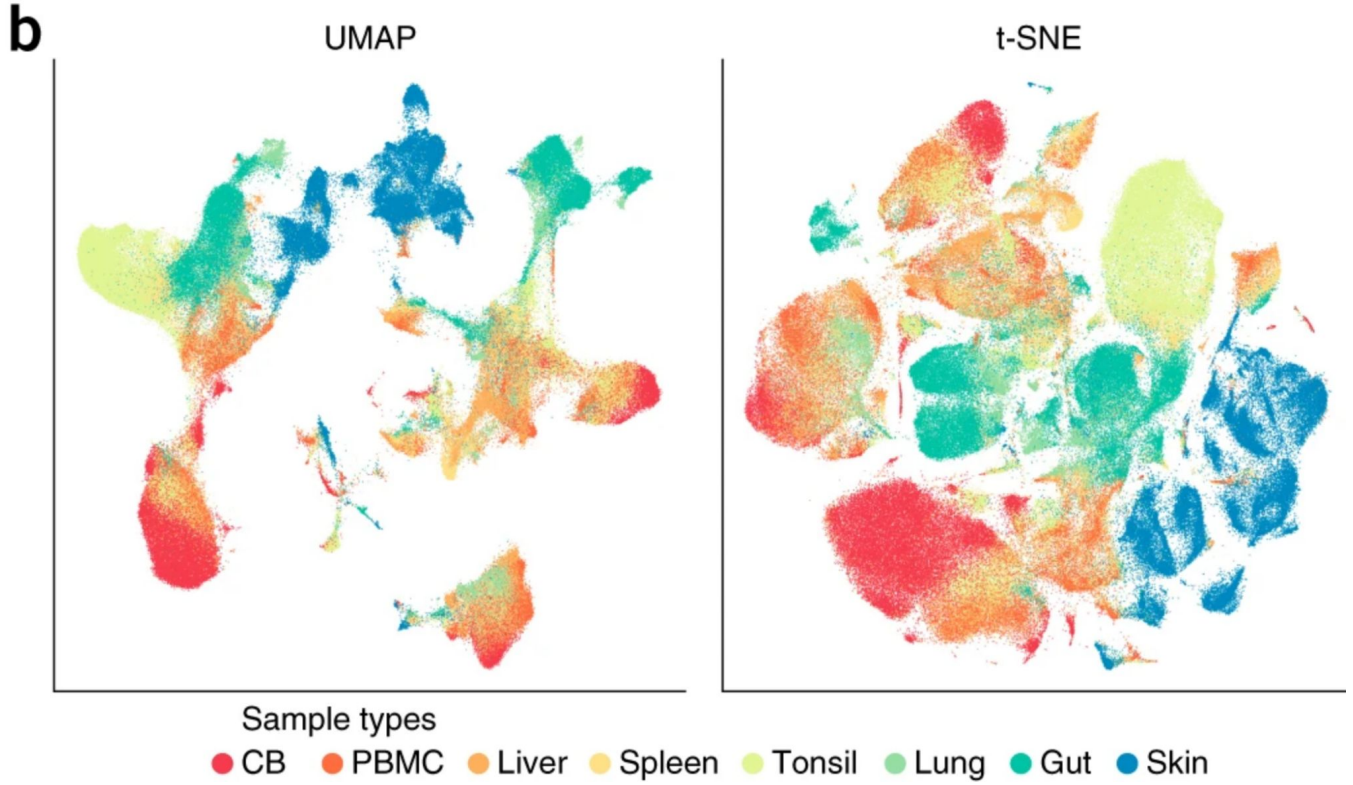
Unsupervised Learning

- Examples of algorithms:
  - **Clustering (day 3 with Mackenzie Tapp)**



Unsupervised Learning

- Examples of algorithms:
  - **Dimensionality reduction**



Measuring Performance (classification)

- Key Performance Indicators (KPI)
  - **Confusion Matrix**
  - Track correct predictions: True Positives (TP) and True Negatives (TN)
  - Track incorrect predictions: False Positives (FP) and False Negatives (FN)

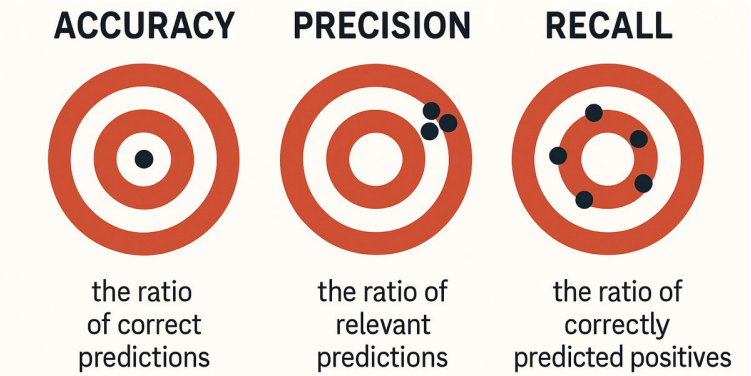
Observed \ Predicted		
	TP	FN
	FP	TN

Problem: *Prediction of having a disease*

TP: *Have* the disease, was predicted as *having*
FP: *Do not* have the disease, was predicted as *having*
FN: *Have* the disease, was predicted as *not having*
TN: *Do not* have the disease, was predicted as *not having*

Measuring Performance (classification)

- Key Performance Indicators (KPI)
  - **Confusion Matrix**
    - **Accuracy** = Overall fraction of correct predictions out of all predictions.
      - Accuracy = (TP+TN)/(TP+TN+FP+FN)
    - **Precision** = Fraction of correct positive predictions
      - Precision = TP/(TP+FP)
    - **Recall (sensitivity)** = Fraction of actual positive cases the model captured.
      - Recall = TP/(TP+FN)
    - **Specificity** = Fraction of actual negative cases the model captured.
      - Specificity = TN/(TN+FP)



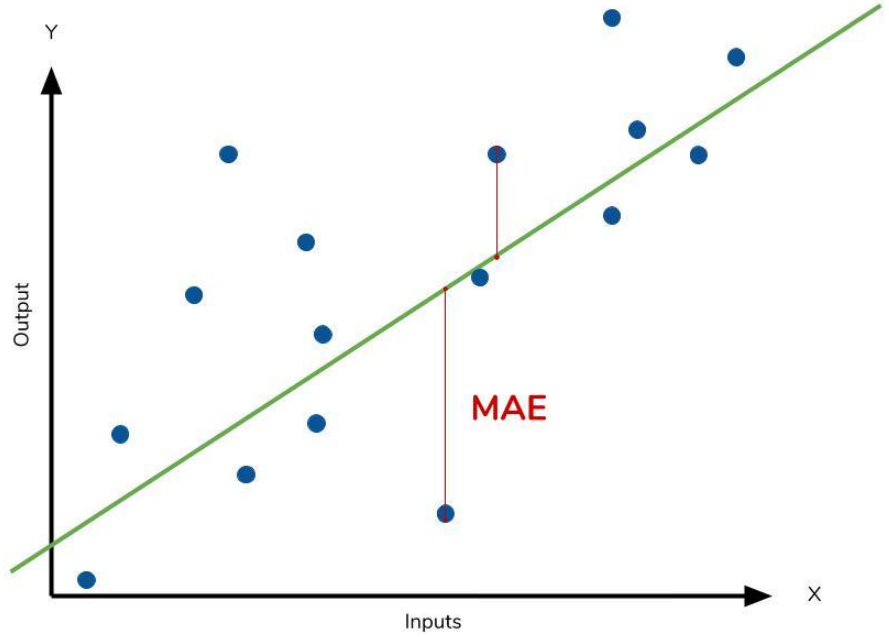
Measuring Performance (Regression)

- We measure the difference between the ground truth targets (actual) and the model's predicted values (predicted).
- These differences are referred to as residuals (error).

Measuring Performance (Regression)

- **Mean Absolute Error (MAE):**
  - *It asks*: how wrong are we on average?
  - Original units as the input data.

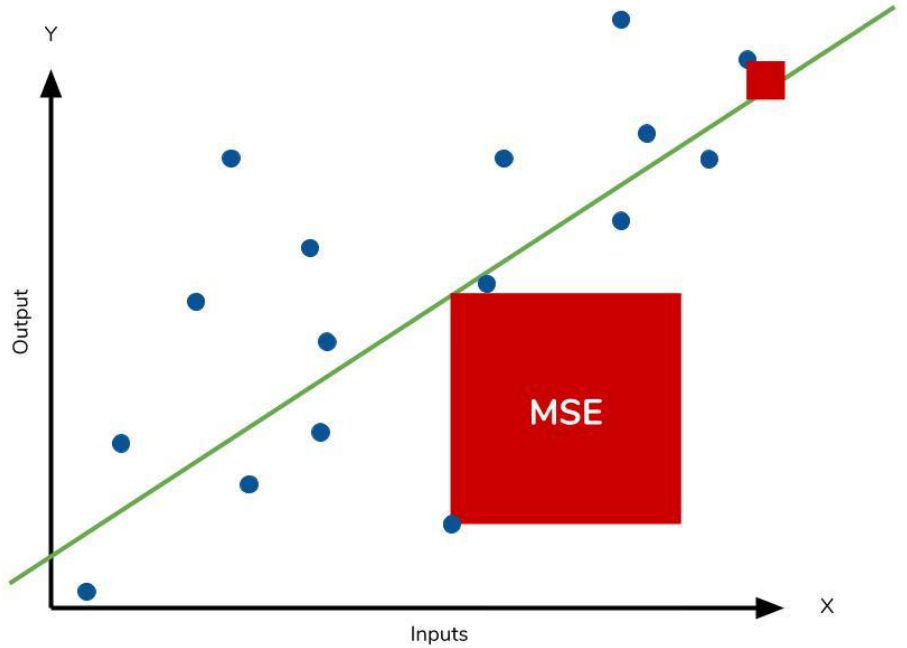
$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} | x_i - x |$$



Measuring Performance (Regression)

- **Root Mean Square Error (RMSE):**
  - *It asks*: how wrong are we on average *if we penalize large errors heavily*?
  - Original units as the input data.

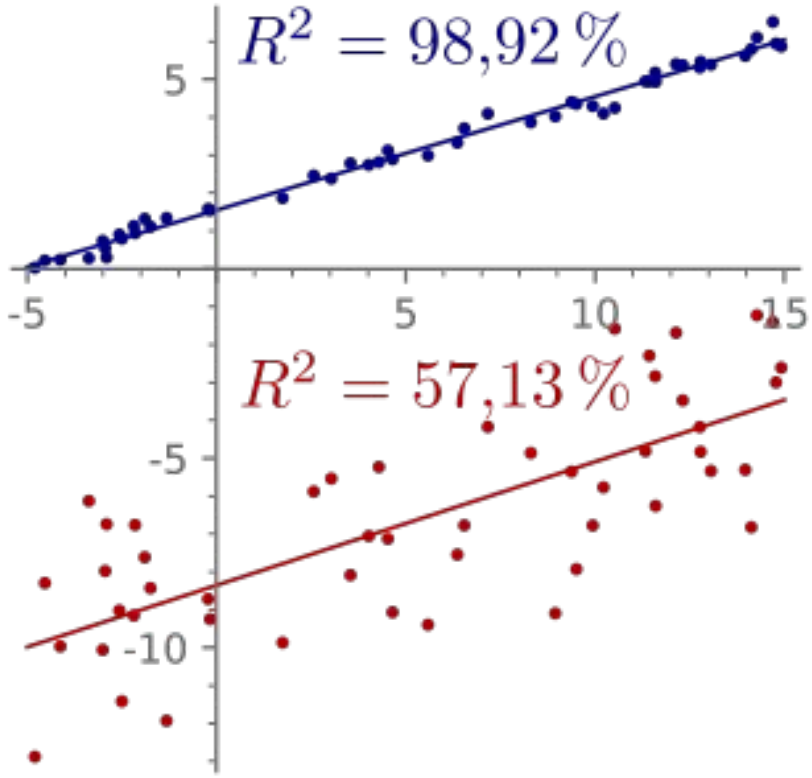
$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(\hat{y}_i - y_i)^2}{n}}$$



Measuring Performance (Regression)

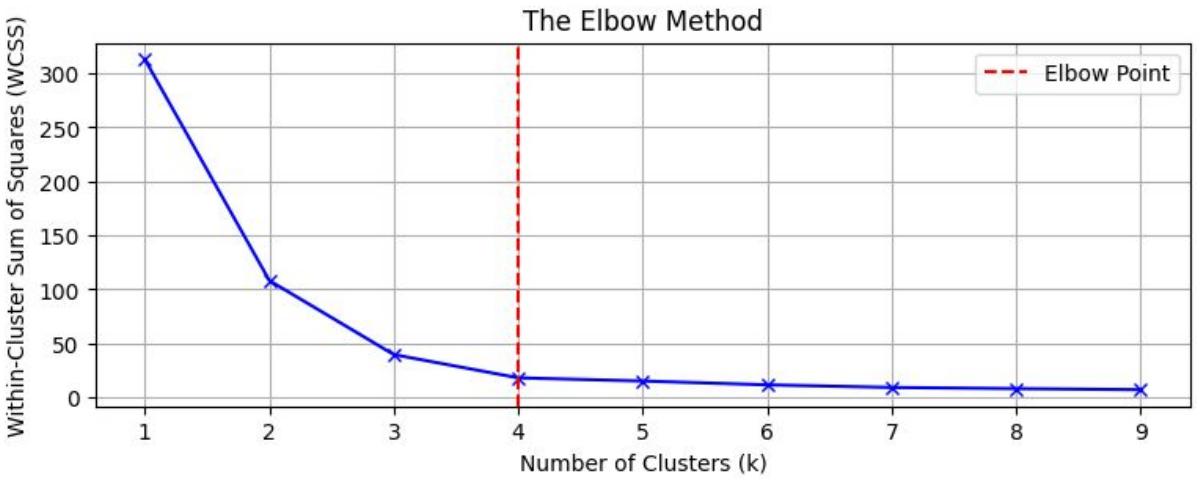
- **Coefficient of Determination (R^2)**
  - *It asks*: how much variation does our model explain?
    - $R^2 \approx 1$: perfect fit; the model explains 100% of the variation.
    - $R^2 \approx 0$: no improvement over predicting the mean.
    - R2 < 0: the predictions are even worse than the class average.

$$R^2 = 1 - \frac{\sum_{i=1}^{m} (X_i - Y_i)^2}{\sum_{i=1}^{m} (\bar{Y} - Y_i)^2}$$



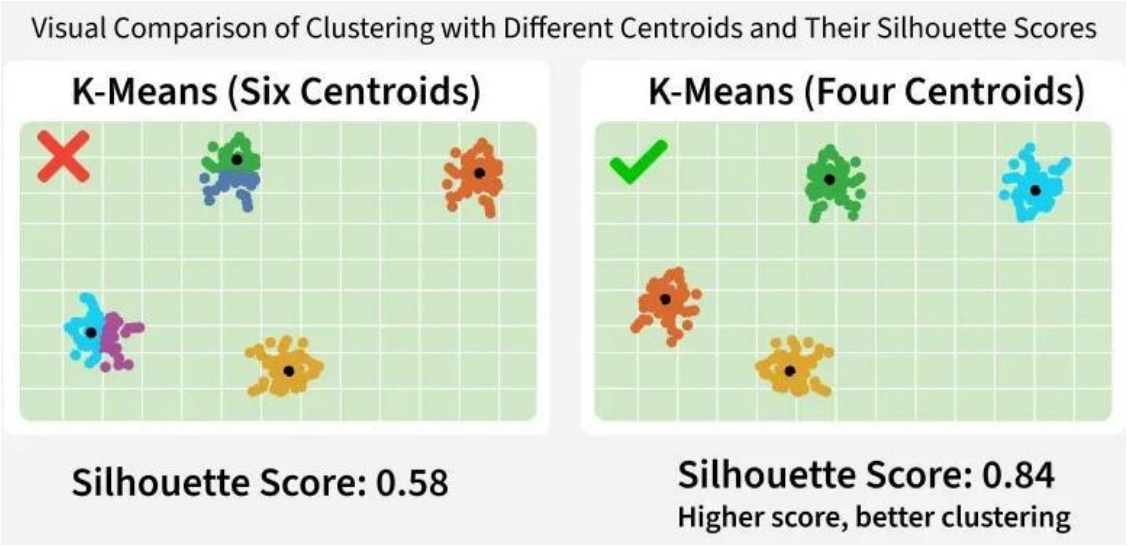
Measuring Performance (Clustering)

- Elbow Method: Find the optimal number of clusters.
- Run a clustering algorithm in a range of k values (number of clusters):
  - For each k, we compute a metric that measures how tight the data points are inside their clusters.



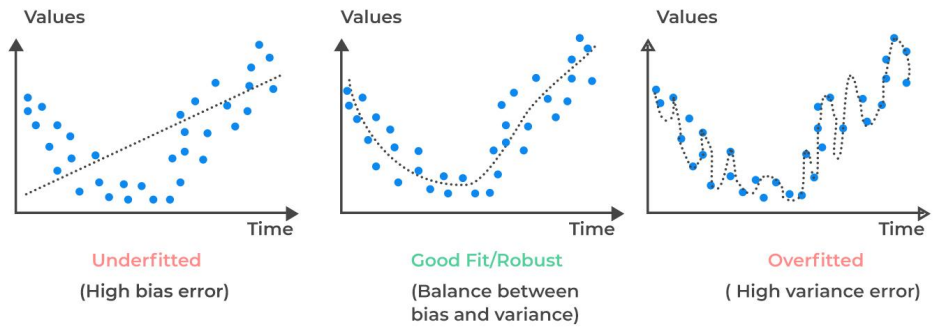
Measuring Performance (Clustering)

- Silhouette Score: Explains the quality of clustering.
- For each clustered data point, we calculate:
  - Cohesion (average distance between a point and all other points of a cluster; smaller score = better clusters).
  - Separation (average distance between a point and all points in the nearest neighboring cluster; larger values = better clusters).



Over/Underfitting

- Scenario: Student training for an exam.
  - **Underfitting (the unprepared student)**
    - The student misses fundamental principles. Scores poorly on practice problems (high training error) and fail the final exam (high test error).
  - **Overfitting (the memorizer)**
    - The student memorizes every question and answer from the practice exam. Scores 100% on practice problems (low training error), but fails when given slightly modified questions (high test error).



Coffee Break & Networking Session

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